



DATA MINING · WEEK 10 GROUP PROJECT

Min-Apriori for Quantitative Association Rule Mining

Dataset 1 · Retail Market Basket Analysis (Discrete Quantities)

min_sup = 3 · 5 transactions · 5 items · 12 rules generated

Hantz 1113534 / Max 1113535

Why Min-Apriori?

Classical Apriori discards magnitude — Min-Apriori preserves it.



Classical Apriori (binary)

Sees only presence/absence:

```
Espresso × 5, Croissant × 1 →  
{1, 1}
```

- A 5-espresso run looks identical to a 1-espresso run.
- Co-occurrence intensity is lost.
- Combo offers built on this miss volume signals.



Min-Apriori (quantitative)

Bounds joint signal by the weakest member:

```
Espresso × 5, Croissant × 1 →  
min = 1
```

- Joint support reflects matched, balanced co-purchase.
- Magnitude carries through to confidence.
- Surfaces actionable bundles for combo design.

The Algorithm in Three Formulas

Min-Apriori = Apriori with the min bounding function.

1. 1-ITEMSET SUPPORT

$$Sup(\{i\}) = \sum_t w_{t,i}$$

Sum of the column across transactions.

2. k-ITEMSET SUPPORT

$$Sup(\{i_1 \dots i_k\}) = \sum_t \min(w_{t,i_1}, \dots, w_{t,i_k})$$

Per transaction, take the minimum across the items, then sum.

3. CONFIDENCE

$$Conf(A \rightarrow B) = \frac{Sup(A \cup B)}{Sup(A)}$$

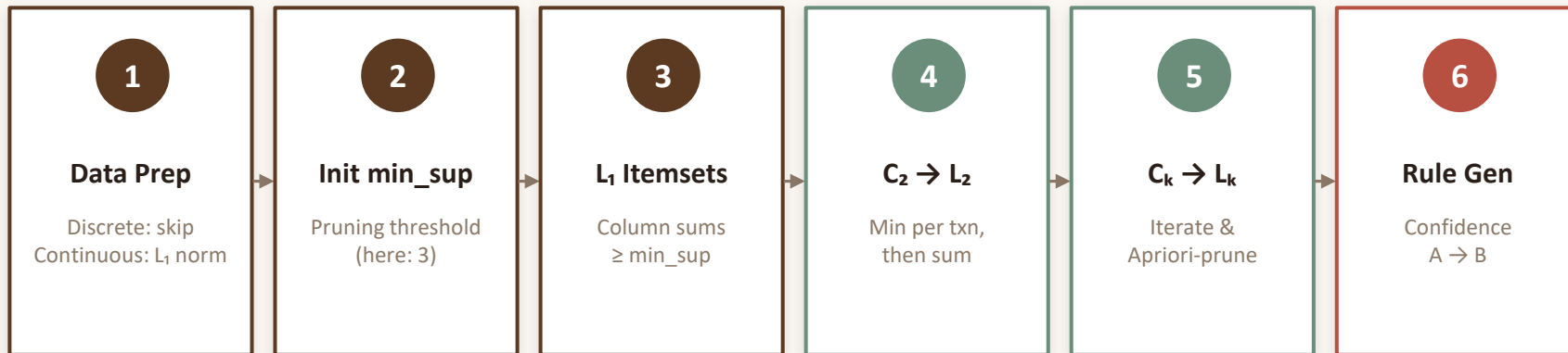
Standard ratio — the directional metric for rule strength.



Apriori property still holds. For $S' \subseteq S$, min over fewer items can only be $\geq$ min over more, so $Sup(S') \geq Sup(S)$. This justifies (k-1)-subset pruning.

Algorithm Flow

Level-wise: each pass either grows the frequent set or terminates.



↻ iterate & prune ↻

■ Setup steps ■ Iterative core ■ Output stage

The Dataset

Five customer baskets at a bakery & coffee shop · cells = quantity purchased.

	Espresso	Croissant	Muffin	Latte	Tea
T1	2	2	0	0	1
T2	0	1	3	2	0
T3	1	2	0	1	2
T4	3	0	2	0	1
T5	2	3	1	1	0

PER-ITEM TOTALS

Espresso **8**

Croissant **8**

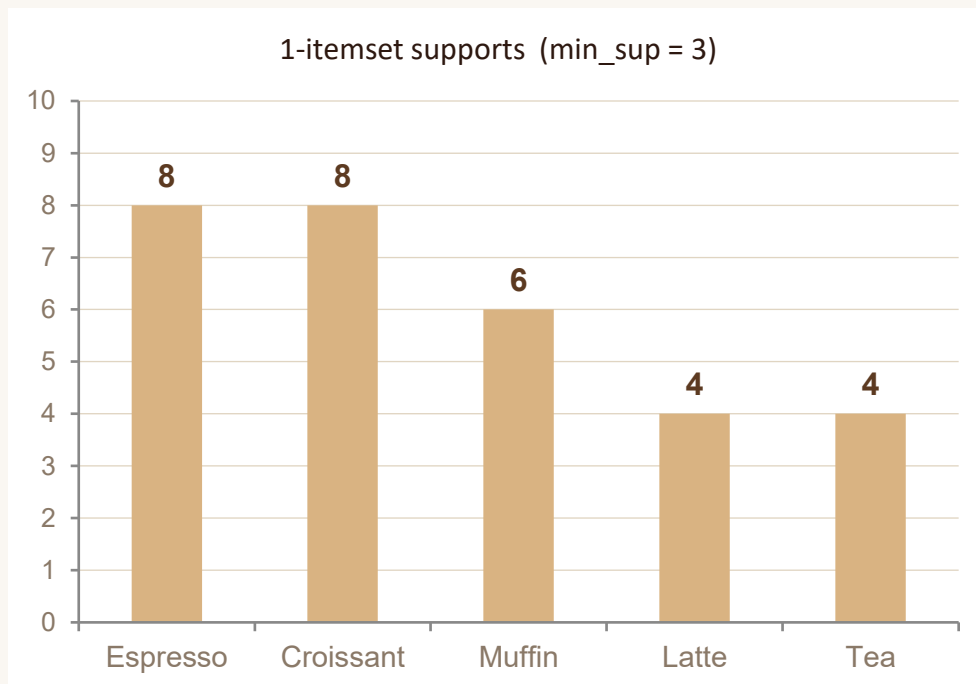
Muffin **6**

Latte **4**

Tea **4**

Step 3 • Frequent 1-Itemsets (L_1)

Sum each column; retain those with support ≥ 3 .



Step 4 • Frequent 2-Itemsets (L₂)

Per transaction, take the min across the pair; sum across transactions.

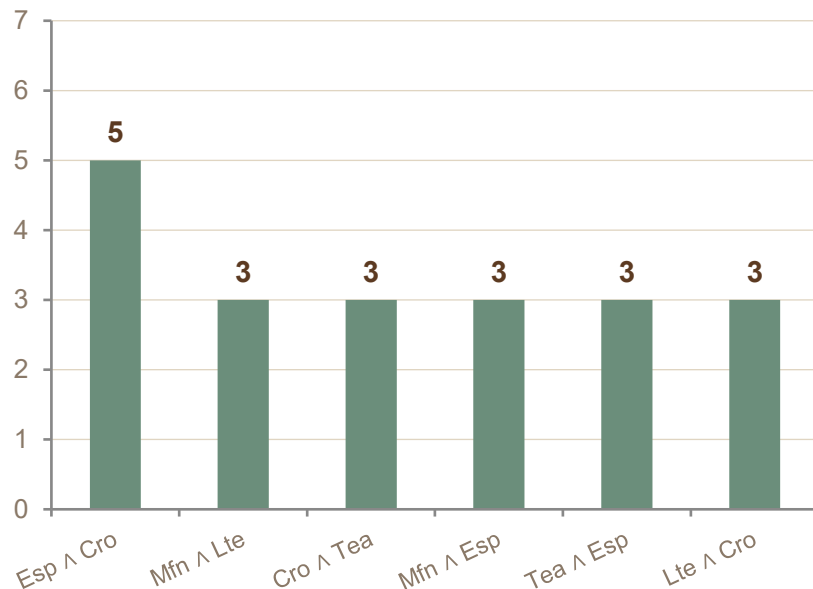
WORKED EXAMPLE

{ Espresso, Croissant }

Txn	min(Esp, Cro)	value
T1	min(2, 2)	2
T2	min(0, 1)	0
T3	min(1, 2)	1
T4	min(3, 0)	0
T5	min(2, 3)	2

$$\text{Sup} = 2 + 0 + 1 + 0 + 2 = 5 \quad \checkmark \geq 3$$

Frequent 2-itemset supports (6 pairs survive of 10)



Candidate	min per txn (T1, T2, T3, T4, T5)	Sup	Verdict
{Esp, Cro}	(2, 0, 1, 0, 2)	5	✓ keep
{Esp, Mfn}	(0, 0, 0, 2, 1)	3	✓ keep
{Esp, Lat}	(0, 0, 1, 0, 1)	2	✗ prune (< 3)
{Esp, Tea}	(1, 0, 1, 1, 0)	3	✓ keep
{Cro, Mfn}	(0, 1, 0, 0, 1)	2	✗ prune
{Cro, Lat}	(0, 1, 1, 0, 1)	3	✓ keep
{Cro, Tea}	(1, 0, 2, 0, 0)	3	✓ keep
{Mfn, Lat}	(0, 2, 0, 0, 1)	3	✓ keep
{Mfn, Tea}	(0, 0, 0, 1, 0)	1	✗ prune
{Lat, Tea}	(0, 0, 1, 0, 0)	1	✗ prune

1. (Cro, Esp)
2. (Cro, Lat)
3. (Cro, Tea)
4. (Esp, Mfn)
5. (Esp, Tea)
6. (Lat, Mfn)

Pairs sharing the first element:

- 1 & 2 (both start with Cro) $\rightarrow \{\text{Cro, Esp, Lat}\}$
- 1 & 3 (both start with Cro) $\rightarrow \{\text{Cro, Esp, Tea}\}$
- 2 & 3 (both start with Cro) $\rightarrow \{\text{Cro, Lat, Tea}\}$
- 4 & 5 (both start with Esp) $\rightarrow \{\text{Esp, Mfn, Tea}\}$

Candidate	2-subsets	All in L_2 ?	Verdict
{Cro, Esp, Lat}	{Cro,Esp} $\checkmark$, {Cro,Lat} $\checkmark$, {Esp,Lat} $\times$	No	$\times$ prune
{Cro, Esp, Tea}	{Cro,Esp} $\checkmark$, {Cro,Tea} $\checkmark$, {Esp,Tea} $\checkmark$	Yes	$\checkmark$ survive
{Cro, Lat, Tea}	{Cro,Lat} $\checkmark$, {Cro,Tea} $\checkmark$, {Lat,Tea} $\times$	No	$\times$ prune
{Esp, Mfn, Tea}	{Esp,Mfn} $\checkmark$, {Esp,Tea} $\checkmark$, {Mfn,Tea} $\times$	No	$\times$ prune

Step 5 · Higher Orders ($L_3 = \emptyset$)

Apriori-prune kills almost everything; the lone survivor fails the support test.

✂ Apriori pruning

Any candidate $\{X, Y, Z\}$ is pruned if any of its 2-subsets is missing from L_2 .

Example: {Espresso, Muffin, Latte}

✂ pruned — {Espresso, Latte} $\notin L_2$

Sole survivor of pruning:

{ Espresso, Croissant, Tea }

FAILS THE SUPPORT TEST

{ Espresso, Croissant, Tea }

$$\begin{aligned}\text{Sup} &= \min(2,2,1) + \min(0,1,0) \\ &\quad + \min(1,2,2) + \min(3,0,1) \\ &\quad + \min(2,3,0) \\ &= 1 + 0 + 1 + 0 + 0 \\ &= 2 < 3 \quad \text{✂}\end{aligned}$$

$L_3 = \emptyset \rightarrow$ algorithm terminates.

Two Highlighted Rules

One picked for largest joint support, one for highest confidence — different stories.

RULE R1 · largest joint support

{ Espresso } → { Croissant }

62.5%

$$\text{conf} = \text{Sup}(\{E, C\}) / \text{Sup}(\{E\}) = 5 / 8$$

Classic breakfast pairing — bought together in matched quantities, suggesting per-person co-purchase.

RULE R2 · highest confidence tier

{ Latte } → { Muffin }

75.0%

$$\text{conf} = \text{Sup}(\{L, M\}) / \text{Sup}(\{L\}) = 3 / 4$$

Fewer Latte buyers than Espresso buyers — but the ones who do are very reliably also buying Muffins.

Min-Apriori vs Classical Apriori

Same dataset, same rules — Min-Apriori rewards quantity-balanced co-occurrence.

Rule	Min-Apriori	Classical	Δ
{Espresso} → {Croissant}	62.5%	62.5%	—
{Latte} → {Muffin}	75.0%	66.7%	+8.3 pp
{Tea} → {Croissant}	75.0%	66.7%	+8.3 pp
{Tea} → {Espresso}	75.0%	66.7%	+8.3 pp

WHEN THEY DIVERGE

Min-Apriori reports higher confidence whenever co-occurring items appear in matched, significant quantities — not just any joint presence.

Better signal for combo-offer design.

Why does {Espresso} → {Croissant} agree? Espresso always co-occurs with comparable Croissant quantities, so the binary projection loses no information here.

Takeaways

Min-Apriori on Dataset 1 · in three sentences.

1

Algorithm worked as designed

$L_1 = 5$ items, $L_2 = 6$ pairs, $L_3 = \emptyset$, 12 rules generated. Manual derivation matches the function output exactly.

2

The min function is the whole idea

Quantitative co-occurrence is bounded by the weakest member. Items bought together in matched amounts get rewarded.

3

Methodology earns its keep

On rules where binary projection loses information, Min-Apriori adds 8.3 pp of confidence — a real difference for combo-offer design.

Thank you · questions?